Chapter 2

Power Calculations

2.1 INTRODUCTION

Power computations are essential in analyzing and designing power electronics circuits. Basic power concepts are reviewed in this chapter, with particular emphasis on power calculations for circuits with non-sinusoidal voltages and currents.

Extra treatment is given to some special cases that are encountered frequently in power electronics.

2.2 Power and Energy

Instantaneous Power

The instantaneous power for any device is computed from the voltage across it and the current in it. *Instantaneous power* is

$$p(t) = v(t)\ i(t) \qquad (2-1)$$

This relationship is valid for any device or circuit. Instantaneous power is generally a time-varying quantity. If the passive sign convention illustrated in Fig. 2-1*a* is observed, the device is absorbing power if $p(t)$ is positive at a specified value of time t. The device is supplying power if $p(t)$ is negative. Sources frequently have an assumed current direction consistent with supplying power. With the convention of Fig. 2-1*b*, a positive $p(t)$ indicates the source is supplying power.

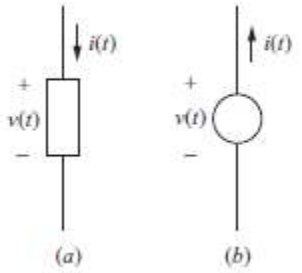


Figure 2-1 (*a*) Passive sign convention: $p(t) > 0$ indicates power is being absorbed; (*b*) $p(t) > 0$ indicates power is being supplied by the source.

Energy

Energy, or work, is the integral of instantaneous power. Observing the passive sign convention, energy absorbed by a component in the time interval from t_1 to t_2 is:

$$W = \int_{t_1}^{t_2} p(t)\,dt \qquad (2\text{-}2)$$

If $v(t)$ is in volts and $i(t)$ is in amperes, power has units of watts and energy has units of joules.

Average Power

Periodic voltage and current functions produce a periodic instantaneous power function. Average power is the time average of $p(t)$ over one or more periods.

Average power P is computed from:

$$P = \frac{1}{T}\int_{t_0}^{t_0+T} p(t)\,dt = \frac{1}{T}\int_{t_0}^{t_0+T} v(t)i(t)\,dt \qquad (2\text{-}3)$$

Where T is the period of the power waveform. Combining Eqs. (2-3) and (2-2), power is also computed from energy per period:

$$P = \frac{W}{T} \qquad (2\text{-}4)$$

Average power is sometimes called *real power* or *active power*, especially in AC circuits. The term *power* usually means average power. The total average power absorbed in a circuit equals the total average power supplied.

EXAMPLE 2-1

Voltage and current, consistent with the passive sign convention, for a device are shown in Fig. 2-2*a* and *b*.

(*a*) Determine the instantaneous power $p(t)$ absorbed by the device.

(*b*) Determine the energy absorbed by the device in one period.

(*c*) Determine the average power absorbed by the device.

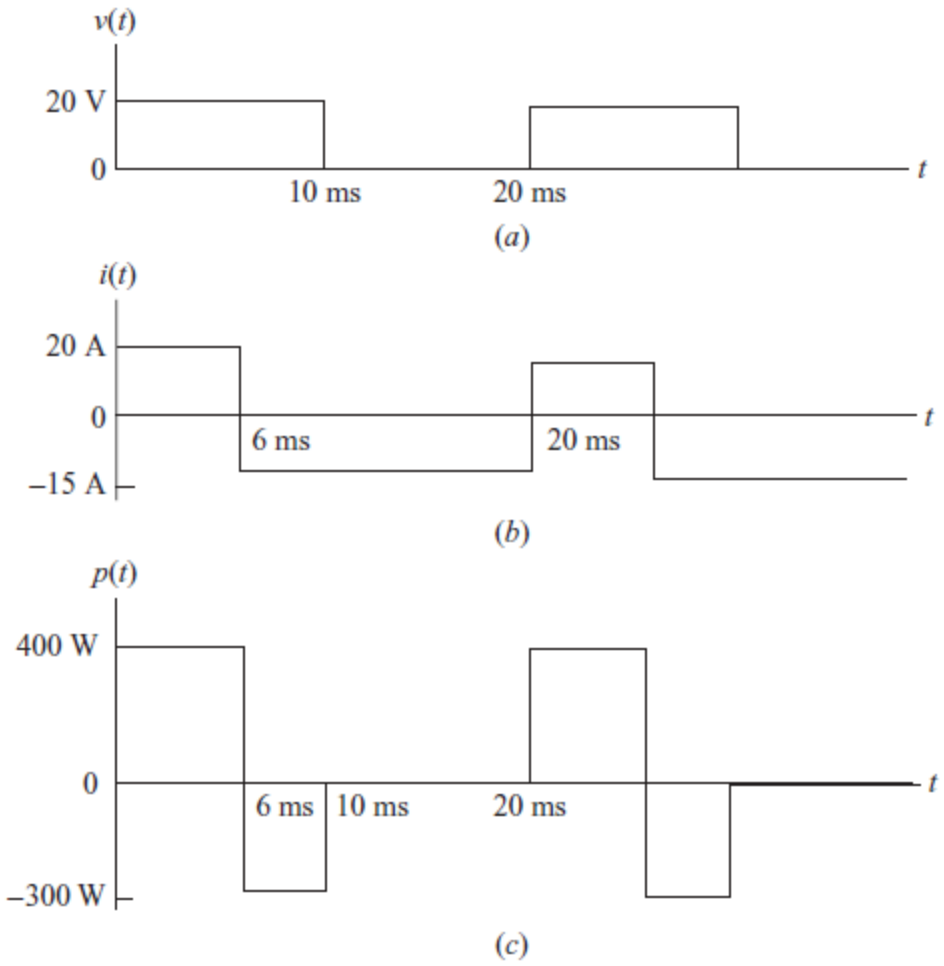


Figure 2-2 Voltage, current, and instantaneous power for Example 2-1.

■ **Solution**

(*a*) The instantaneous power is computed from Eq. (2-1). The voltage and current are expressed as:

$$v(t) = \begin{cases} 20\text{ V} & 0 < t < 10\text{ ms} \\ 0 & 10\text{ ms} < t < 20\text{ ms} \end{cases}$$

$$i(t) = \begin{cases} 20\text{ V} & 0 < t < 6\text{ ms} \\ -15\text{ A} & 6\text{ ms} < t < 20\text{ ms} \end{cases}$$

Instantaneous power, shown in Fig. 2-2*c*, is the product of voltage and current and is expressed as:

$$p(t) = \begin{cases} 400\ \text{W} & 0 < t < 6\ \text{ms} \\ -300\ \text{W} & 6\ \text{ms} < t < 10\ \text{ms} \\ 0 & 10\ \text{ms} < t < 20\ \text{ms} \end{cases}$$

(*b*) Energy absorbed by the device in one period is determined from Eq. (2-2).

$$W = \int_0^T p(t)\,dt = \int_0^{0.006} 400\,dt + \int_{0.006}^{0.010} -300\,dt + \int_{0.010}^{0.020} 0\,dt = 2.4 - 1.2 = 1.2\ \text{J}$$

(*c*) Average power is determined from Eq. (2-3).

$$P = \frac{1}{T}\int_0^T p(t)dt = \frac{1}{0.020}\left(\int_0^{0.006} 400\,dt + \int_{0.006}^{0.010} -300\,dt + \int_{0.010}^{0.020} 0\,dt\right)$$

$$= \frac{2.4 - 1.2 - 0}{0.020} = 60\ \text{W}$$

Average power could also be computed from Eq. (2-4) by using the energy per period from part (b).

$$P = \frac{W}{T} = \frac{1.2\ \text{J}}{0.020\ \text{s}} = 60\ \text{W}$$

A special case that is frequently encountered in power electronics is the power absorbed or supplied by a dc source. Applications include battery-charging circuits and dc power supplies. The average power absorbed by a dc voltage source $v(t) = V_{dc}$ that has a periodic current $i(t)$ is derived from the basic definition of average power in Eq. (2-3):

$$P_{\text{dc}} = \frac{1}{T}\int_{t_0}^{t_0+T} v(t)i(t)\,dt = \frac{1}{T}\int_{t_0}^{t_0+T} V_{\text{dc}}\,i(t)\,dt$$

Bringing the constant V_{dc} outside of the integral gives:

$$P_{\text{dc}} = V_{\text{dc}}\left[\frac{1}{T}\int_{t_0}^{t_0+T} i(t)\,dt\right]$$

The term in brackets is the average of the current waveform. Therefore, *average power absorbed by a DC voltage source is the product of the voltage and the average current.*

$$\boxed{P_{\text{dc}} = V_{\text{dc}}\, I_{\text{avg}}}$$

Similarly, average power absorbed by a DC source $i(t) = I_{dc}$ is:

$$P_{\text{dc}} = I_{\text{dc}}\, V_{\text{avg}}$$

2.3 Inductors and Capacitors

Inductors and capacitors have some particular characteristics that are important in power electronics applications. For periodic currents and voltages,

$$\begin{aligned} i(t+T) &= i(t) \\ v(t+T) &= v(t) \end{aligned} \qquad (2\text{-}7)$$

For an inductor, the stored energy is:

$$w(t) = \frac{1}{2} L i^2(t) \qquad (2\text{-}8)$$

If the inductor current is periodic, the stored energy at the end of one period is the same as at the beginning. No net energy transfer indicates that *the average power absorbed by an inductor is zero for steady-state periodic operation.*

$$\boxed{P_L = 0} \qquad (2\text{-}9)$$

Instantaneous power is not necessarily zero because power may be absorbed during part of the period and returned to the circuit during another part of the period. Furthermore, from the voltage-current relationship for the inductor

$$i(t_0 + T) = \frac{1}{L} \int_{t_0}^{t_0+T} v_L(t)\, dt + i(t_0) \qquad (2\text{-}10)$$

Rearranging and recognizing that the starting and ending values are the same for periodic currents, we have:

$$i(t_0 + T) - i(t_0) = \frac{1}{L} \int_{t_0}^{t_0+T} v_L(t)\, dt = 0 \qquad (2\text{-}11)$$

Multiplying by *L*/*T* yields an expression equivalent to the average voltage across the inductor over one period.

$$\text{avg}[v_L(t)] = V_L = \frac{1}{T}\int_{t_0}^{t_0+T} v_L(t)\,dt = 0 \qquad (2\text{-}12)$$

Therefore, *for periodic currents, the average voltage across an inductor is zero.* This is very important and will be used in the analysis of many circuits, including DC-DC converters and DC power supplies.

For a capacitor, stored energy is:

$$w(t) = \frac{1}{2}Cv^2(t) \qquad (2\text{-}13)$$

If the capacitor voltage is periodic, the stored energy is the same at the end of a period as at the beginning. Therefore, *the average power absorbed by the capacitor is zero for steady-state periodic operation.*

$$P_C = 0 \qquad (2\text{-}14)$$

From the voltage-current relationship for the capacitor,

$$v(t_0 + T) = \frac{1}{C}\int_{t_0}^{t_0+T} i_C(t)\,dt + v(t_0) \qquad (2\text{-}15)$$

Rearranging the preceding equation and recognizing that the starting and ending values are the same for periodic voltages, we get

$$v(t_0 + T) - v(t_0) = \frac{1}{C}\int_{t_0}^{t_0+T} i_C(t)\,dt = 0 \qquad (2\text{-}16)$$

Multiplying by *C*/*T* yields an expression for the average current in the capacitor over one period.

$$\text{avg}\,[i_C(t)] = I_C = \frac{1}{T}\int_{t_0}^{t_0+T} i_C(t)\,dt = 0 \qquad (2\text{-}17)$$

Therefore, *for periodic voltages, the average current in a capacitor is zero.*

Example 2-2 (Power and Voltage for an Inductor)
The current in a 5-mH inductor of Fig. 2-3*a* is the periodic triangular wave shown in Fig. 2-3*b*. Determine the voltage, instantaneous power, and average power for the inductor.

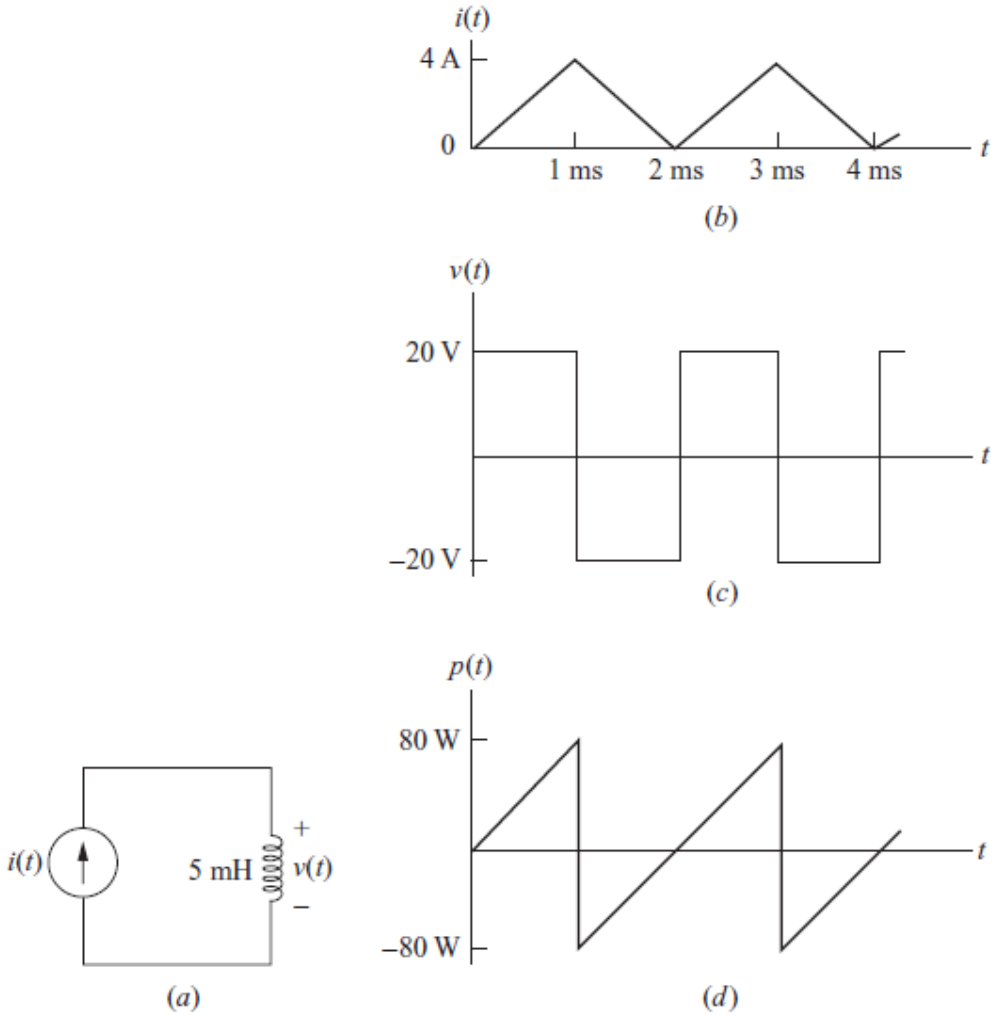


Figure 2.3 (*a*) Circuit for Example 2-2; (*b*) inductor current; (*c*) inductor voltage; (*d*) inductor instantaneous power.

■ Solution

The voltage across the inductor is computed from $v(t) = L(di/dt)$ and is shown in Fig. 2-3*c*. The average inductor voltage is zero, as can be determined from Fig. 2-3*c* by inspection. The instantaneous power in the inductor is determined from $p(t) = v(t)i(t)$ and is shown in Fig. 2-3*d*. When $p(t)$ is positive, the inductor is absorbing power, and when $p(t)$ is

negative, the inductor is supplying power. The average inductor power is zero.

2.4 ENERGY RECOVERY

Inductors and capacitors must be energized and deenergized in several applications of power electronics. For example, a fuel injector solenoid in an automobile is energized for a set time interval by a transistor switch. Energy is stored in the solenoid's inductance when current is established. The circuit must be designed to remove the stored energy in the inductor while preventing damage to the transistor when it is turned off. Circuit efficiency can be improved if stored energy can be transferred to the load or to the source rather than dissipated in circuit resistance. The concept of recovering stored energy is illustrated by the circuits described in this section.

Fig. 2-4*a* shows an inductor that is energized by turning on a transistor switch. The resistance associated with the inductance is assumed to be negligible, and the transistor switch and diode are assumed to be ideal. The diode resistor path provides a means of opening the switch and removing the stored energy in the inductor when the transistor turns off. Without the diode-resistor path, the transistor could be destroyed when it is turned off because a rapid decrease in inductor current would result in excessively high inductor and transistor voltages.

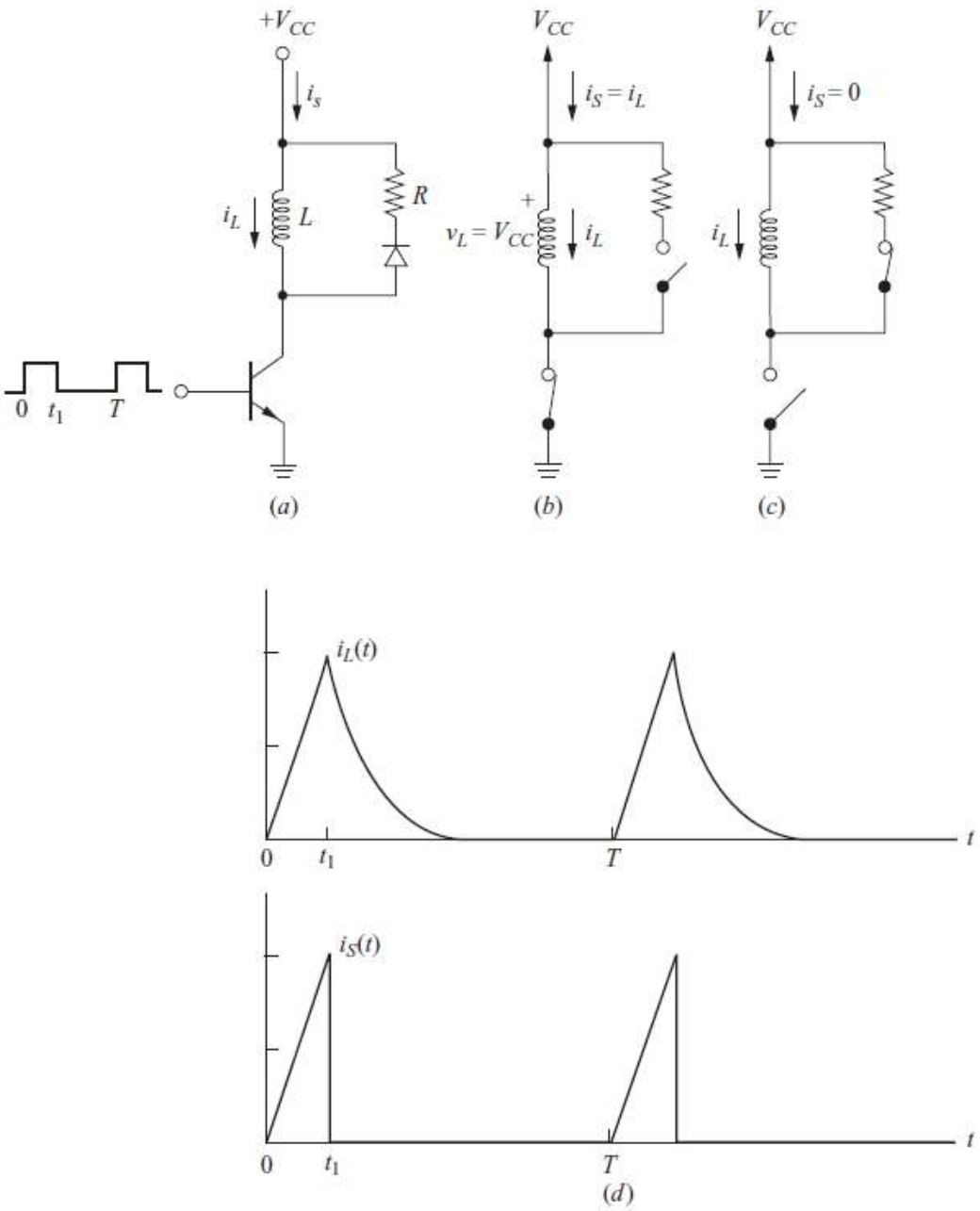


Figure 2-4 (*a*) A circuit to energize an inductance and then transfer the stored energy to a resistor; (*b*) Equivalent circuit when the transistor is on; (*c*) Equivalent circuit when the transistor is off, and the diode is on; (*d*) Inductor and source currents.

Assume that the transistor switch turns on at $t = 0$ and turns off at $t = t_1$. The circuit is analyzed first for the transistor switch on and then for the switch off.

Transistor on: $0 < t < t_1$

The voltage across the inductor is V_{CC}, and the diode is reverse-biased when the transistor is on (Fig. 2-4*b*).

$$v_L = V_{CC} \qquad (2\text{-}18)$$

An expression for inductor current is obtained from the voltage-current relationship:

$$i_L(t) = \frac{1}{L}\int_0^t v_L(\lambda)\, d\lambda + i_L(0) = \frac{1}{L}\int_0^t V_{CC}\, d\lambda + 0 = \frac{V_{CC}t}{L} \qquad (2\text{-}19)$$

The source current is the same as the inductor current.

$$i_s(t) = i_L(t) \qquad (2\text{-}20)$$

Inductor and source currents thus increase linearly when the transistor is on. The circuit is next analyzed for the transistor switch-off.

Transistor off: $t_1 < t < T$

In the interval $t_1 < t < T$, the transistor switch is off and the diode is on (Fig. 2-4*c*).

The current in the source is zero, and the current in the inductor and resistor is a decaying exponential with time constant *L*/*R*. The initial condition for inductor current is determined from Eq. (2-19):

$$i_L(t_1) = \frac{V_{CC}t_1}{L} \qquad (2\text{-}21)$$

Inductor current is then expressed as

$$i_L(t) = i_L(t_1)e^{-(t-t_1)/\tau} = \left(\frac{V_{CC}t_1}{L}\right)e^{-(t-t_1)/\tau} \quad t_1 < t < T \qquad (2\text{-}22)$$

Where $\tau = \frac{L}{R}$

Source current is zero when the transistor is off.

$$i_S = 0 \qquad (2\text{-}23)$$

Average power supplied by the DC source during the switching period is determined from the product of voltage and average current [Eq. (2-5)].

$$P_S = V_S I_S = V_{CC}\left[\frac{1}{T}\int_0^T i_s(t)\,dt\right]$$

$$= V_{CC}\left[\frac{1}{T}\int_0^{t_1}\frac{V_{CC}t}{L}\,dt + \frac{1}{T}\int_{t_1}^{T} 0\,dt\right] = \frac{(V_{CC}t_1)^2}{2LT} \qquad (2\text{-}24)$$

Average power absorbed by the resistor could be determined by integrating an expression for instantaneous resistor power, but an examination of the circuit reveals an easier way. The average power absorbed by the inductor is zero, and power absorbed by the ideal transistor and diode is zero. Therefore, all power supplied by the source must be absorbed by the resistor:

$$P_R = P_S = \frac{(V_{CC}t_1)^2}{2LT} \qquad (2\text{-}25)$$

Another way to approach the problem is to determine the peak energy stored in the inductor,

$$W = \frac{1}{2}Li^2(t_1) = \frac{1}{2}L\left(\frac{V_{CC}t_1}{L}\right)^2 = \frac{(V_{CC}t_1)^2}{2L} \qquad (2\text{-}26)$$

The energy stored in the inductor is transferred to the resistor while the transistor switch is open. Power absorbed by the resistor can be determined from Eq. (2-4).

$$P_R = \frac{W}{T} = \frac{(V_{CC}t_1)^2}{2LT} \qquad (2\text{-}27)$$

which must also be the power supplied by the source. The function of the resistor in this circuit of Fig. 2-4*a* is to absorb the stored energy in the inductance and protect the transistor. This energy is converted to heat and represents a power loss in the circuit.

Another way to remove the stored energy in the inductor is shown in Fig. 2-5*a*. Two transistor switches are turned on and off simultaneously. The diodes provide a means of returning energy stored in the inductor back to the source. Assume that the transistors turn on at $t = 0$ and turn off at $t = t_1$. The analysis of the circuit of Fig. 2-5*a* begins with the transistors on.

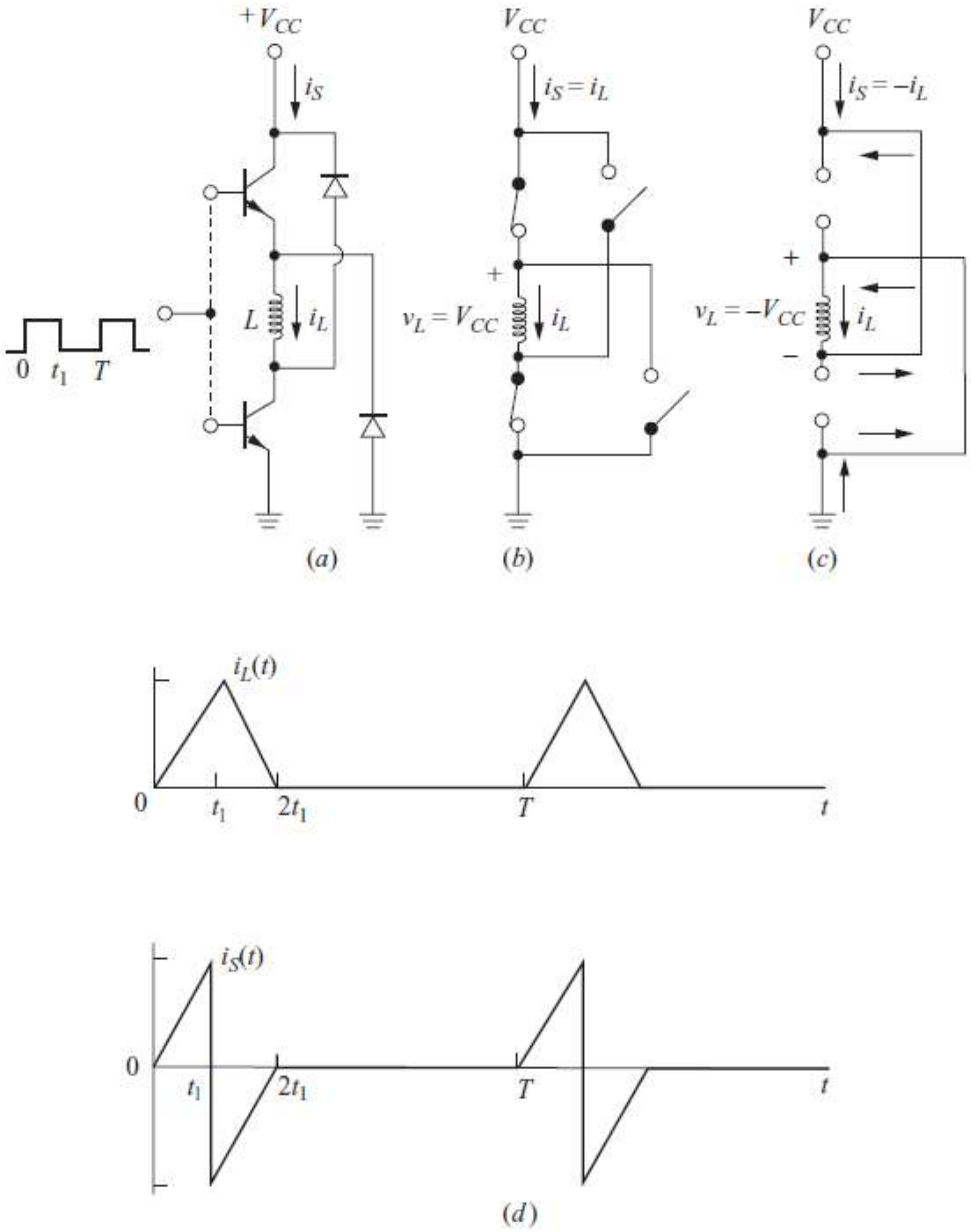


Figure 2-5 (*a*) Acircuit to energize an inductance and recover the stored energy by transferring it back to the source; (*b*) Equivalent circuit when the transistors are on; (*c*) Equivalent circuit when the transistors are off and the diodes are on; (*d*) Inductor and source currents.

Transistors on: $0 < t < t_1$

When the transistors are on, the diodes are reverse-biased, and the voltage across the inductor is V_{CC}. The inductor voltage is the same as the source when the transistors are on (Fig. 2-5*b*):

$$v_L = V_{CC} \qquad (2\text{-}28)$$

Inductor current is the function

$$i_L(t) = \frac{1}{L}\int_0^t v_L(\lambda)\, d\lambda + i_L(0) = \frac{1}{L}\int_0^t V_{CC}\, d\lambda + 0 = \frac{V_{CC}t}{L} \qquad (2\text{-}29)$$

Source current is the same as inductor current.

$$i_S(t) = i_L(t) \qquad (2\text{-}30)$$

From the preceding equations, inductor and source currents increase linearly while the transistor switches are on, as was the case for the circuit of Fig. 2-4*a*.

The circuit is next analyzed for the transistors off.

Transistors off: $t_1 < t < T$

When the transistors are turned off, the diodes become forward-biased to provide a path for the inductor current (Fig. 2-5*c*). The voltage across the inductor then becomes the opposite of the source voltage:

$$v_L = -V_{CC} \qquad (2\text{-}31)$$

An expression for inductor current is obtained from the voltage-current relationship.

$$i_L(t) = \frac{1}{L}\int_{t_1}^t v_L(\lambda)\, d\lambda + i_L(t_1) = \frac{1}{L}\int_{t_1}^t (-V_{CC})\, d\lambda + \frac{V_{CC}t_1}{L}$$

$$= \left(\frac{V_{CC}}{L}\right)[(t_1 - t) + t_1]$$

or,

$$i_L(t) = \left(\frac{V_{CC}}{L}\right)(2t_1 - t) \qquad t_1 < t < 2t_1 \qquad (2\text{-}32)$$

Inductor current decreases and becomes zero at $t = 2t_1$, at which time the diodes turn off. Inductor current remains at zero until the transistors turn on again.

Source current is the opposite of inductor current when the transistors are off and the diodes are on:

$$i_S(t) = -i_L(t) \qquad (2\text{-}33)$$

The source is absorbing power when the source current is negative. Average source current is zero, resulting in an average source power of zero.

The source supplies power while the transistors are on, and the source absorbs power while the transistors are off and the diodes are on. Therefore, the energy stored in the inductor is recovered by transferring it back to the source. Practical solenoids or other magnetic devices have equivalent resistances that represent losses or energy absorbed to do work, so not all energy will be returned to the source. The circuit of Fig. 2-5*a* has no energy losses inherent to the design and is therefore more efficient than that of Fig. 2-4*a*.

Example 2-3 (Energy Recovery)
The circuit of Fig. 2-4*a* has $V_{CC} = 90\ V$, $L = 200\ mH$, $R = 20\ \Omega$, $t_1 = 10\ ms$, and $T = 100\ ms$.
Determine (*a*) the peak current and peak energy storage in the inductor,
(*b*) the average power absorbed by the resistor.
(*c*) the peak and average power supplied by the source.
(*d*) Compare the results with what would happen if the inductor were energized using the circuit of Fig. 2-5*a*.

■ **Solution**

(*a*) From Eq. (2-19), when the transistor switch is on, inductor current is

$$i_L(t) = \left(\frac{V_{CC}}{L}\right)t = \left(\frac{90}{0.2}\right)t = 450t \text{ A} \qquad 0 < t < 10 \text{ ms}$$

Peak inductor current and stored energy are

$$i_L(t_1) = 450(0.01) = 4.5 \text{ A}$$

$$W_L = \frac{1}{2}Li^2(t_1) = \frac{1}{2}(0.2)(4.5)^2 = 2.025 \text{ J}$$

(*b*) The time constant for the current when the switch is open is
$\frac{L}{R} = \frac{200\ mH}{20\ \Omega} = 10\ ms$.
The switch is open for 90 ms, which is 10 time constants, so essentially all stored energy in the inductor is transferred to the resistor:

$$W_R = W_L = 2.025 \text{ J}$$

Average power absorbed by the resistor is determined from Eq. (2-4):

$$P_R = \frac{W_R}{T} = \frac{2.025 \text{ J}}{0.1 \text{ s}} = 20.25 \text{ W}$$

(*c*) The source current is the same as the inductor current when the switch is closed and is zero when the switch is open. Instantaneous power supplied by the source is

$$p_S(t) = v_S(t)i_S(t) = \begin{cases} (90\text{ V})(450t\text{ A}) = 40{,}500t\text{ W} & 0 < t < 10\text{ ms} \\ 0 & 10\text{ ms} < t < 100\text{ ms} \end{cases}$$

Which has a maximum value of $405\ Wat\ t = \ 10\ ms$. Average power supplied by the source can be determined from Eq. (2-3):

$$P_S = \frac{1}{T}\int_0^T p_S(t)\,dt = \frac{1}{0.1}\left(\int_0^{0.01} 40{,}500t\,dt + \int_{0.01}^{0.1} 0\,dt\right) = 20.25\text{ W}$$

Average source power also can be determined from Eq. (2-5).
Average of the triangular source current waveform over one period is

$$I_S = \frac{1}{2}\left[\frac{(0.01\text{ s})(4.5\text{ A})}{0.1\text{ s}}\right] = 0.225\text{ A}$$

$$I_S = \frac{1}{2}\left[\frac{(0.01\text{ s})(4.5\text{ A})}{0.1\text{ s}}\right] = 0.225\text{ A}$$

and average source power is then

$$P_S = V_{CC}I_S = (90\text{ V})(0.225\text{ A}) = 20.25\text{ W}$$

Still another computation of average source power comes from recognizing that the power absorbed by the resistor is the same as that supplied by the source.

$$P_S = P_R = 20.25\text{ W}$$

(*d*) When the inductor is energized from the circuit of Fig. 2-5*a*, the inductor current is described by Eqs. (2-29) and (2-32).

$$i_L(t) = \begin{cases} 450t\text{ A} & 0 < t < 10\text{ ms} \\ 9 - 450t\text{ A} & 10\text{ ms} < t < 20\text{ ms} \\ 0 & 20\text{ ms} < t < 100\text{ ms} \end{cases}$$

The peak current and peak energy storage are the same as for the circuit of Fig. 2-4*a*.
The source current has the form shown in Fig. 2-5*d* and is expressed as

$$i_S(t) = \begin{cases} 450t\text{ A} & 0 < t < 10\text{ ms} \\ 450t - 9\text{ A} & 10\text{ ms} < t < 20\text{ ms} \\ 0 & 20\text{ ms} < t < 100\text{ ms} \end{cases}$$

Instantaneous power supplied by the source is

$$p_S(t) = 90 i_S(t) = \begin{cases} 40{,}500t \text{ W} & 0 < t < 10 \text{ ms} \\ 40{,}500t - 810 \text{ W} & 10 \text{ ms} < t < 20 \text{ ms} \\ 0 & 20 \text{ ms} < t < 100 \text{ ms} \end{cases}$$

Average source current is zero, and average source power is zero. Peak source power is peak current times voltage, which is 405 Was in part (*c*).

2.5 Effective Values: rms

The effective value of a voltage or current is also known as the root-mean-square (rms) value. The effective value of a periodic voltage waveform is based on the average power delivered to a resistor. For a dc voltage across a resistor,

$$P = \frac{V_{dc}^2}{R} \qquad (2\text{-}34)$$

For a periodic voltage across a resistor, *effective voltage* is defined as the voltage that is as effective as the dc voltage in supplying average power. Effective voltage can be computed using the equation

$$P = \frac{V_{\text{eff}}^2}{R} \qquad (2\text{-}35)$$

Computing average resistor power from Eq. (2-3) gives

$$P = \frac{1}{T}\int_0^T p(t)\,dt = \frac{1}{T}\int_0^T v(t)i(t)\,dt = \frac{1}{T}\int_0^T \frac{v^2(t)}{R}\,dt$$
$$= \frac{1}{R}\left[\frac{1}{T}\int_0^T v^2(t)\,dt\right] \qquad (2\text{-}36)$$

Equating the expressions for average power in Eqs. (2-35) and (2-36) gives

$$P = \frac{V_{\text{eff}}^2}{R} = \frac{1}{R}\left[\frac{1}{T}\int_0^T v^2(t)\,dt\right]$$

or

$$V_{\text{eff}}^2 = \frac{1}{T}\int_0^T v^2(t)\,dt$$

Resulting in the expression for effective or rms voltage

$$V_{\text{eff}} = V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt} \qquad (2\text{-}37)$$

The effective value is the square *root* of the *mean* of the *square* of the voltage— hence the term *root mean square*.
Similarly, rms current is developed from $P = I_{rms}^2$ as:

$$I_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T i^2(t)\,dt} \qquad (2\text{-}38)$$

The usefulness of the rms value of voltages and currents lies in the computing power absorbed by resistances. Additionally, AC power system voltages and currents are invariably given in rms values. Ratings of devices such as transformers are often specified in terms of rms voltage and current.

Example 2-4 (rms Value of a Pulse Waveform)
Determine the rms value of the periodic pulse waveform that has a duty ratio of *D* as shown in Fig. 2-6.

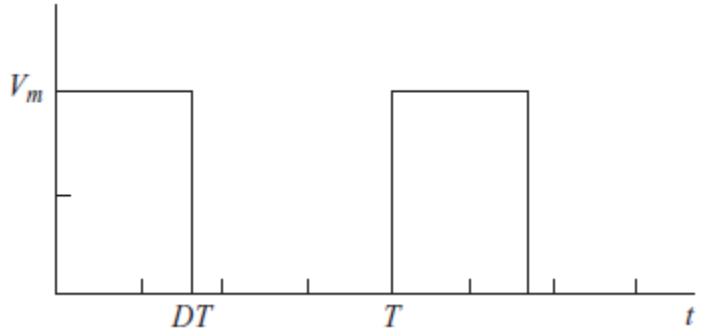


Figure 2-6 Pulse waveform for Example 2-4.

■ **Solution**

The voltage is expressed as

$$v(t) = \begin{cases} V_m & 0 < t < DT \\ 0 & DT < t < T \end{cases}$$

Using Eq. (2-37) to determine the rms value of the waveform gives

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt} = \sqrt{\frac{1}{T}\left(\int_0^{DT} V_m^2\,dt + \int_{DT}^T 0^2 dt\right)} = \sqrt{\frac{1}{T}(V_m^2\,DT)}$$

Yielding

$$V_{\text{rms}} = V_m\sqrt{D}$$

Example 2-5 (rms Values of Sinusoids)

Determine the rms values of :

(a) a sinusoidal voltage of $v(t) = Vm\ sin\ (\omega t)$.

(b) a full-wave rectified sine wave of $v(t) = |Vm\ sin(\omega t)|$

(c) a half-wave rectified sine wave of $v(t) = Vm\ sin(\omega t)$ for $0\ <\ t\ <\ T/2$ and zero otherwise.

■ **Solution**

(a) The rms value of the sinusoidal voltage is computed from Eq. (2-37):

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T V_m^2 \sin^2(\omega t)\,dt} \qquad \text{where } T = \frac{2\pi}{\omega}$$

An equivalent expression uses (ωt)as the variable of integration. Without showing the details of the integration, the result is:

$$V_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} V_m^2 \sin^2(\omega t)\,d(\omega t)} = \frac{V_m}{\sqrt{2}}$$

Note that the rms value is independent of the frequency.

(b) Equation (2-37) can be applied to the full-wave rectified sinusoid, but the results of part (a) can also be used to advantage. The rms formula uses the integral of the square of the function. The square of the sine wave is

identical to the square of the full-wave rectified sine wave, so the rms values of the two waveforms are identical:

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$$

(*c*) Equation (2-37) can be applied to the half-wave rectified sinusoid.

$$V_{\text{rms}} = \sqrt{\frac{1}{2\pi}\left(\int_0^{\pi} V_m^2 \sin^2(\omega t)\, d(\omega t) + \int_{\pi}^{2\pi} 0^2\, d(\omega t)\right)} = \sqrt{\frac{1}{2\pi}\int_0^{\pi} V_m^2 \sin^2(\omega t)\, d(\omega t)}$$

The result of part (*a*) will again be used to evaluate this expression. The square of the function has one-half the area of that of the functions in (*a*) and (*b*).
That is,

$$V_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_0^{\pi} V_m^2 \sin^2(\omega t)\, d(\omega t)} = \sqrt{\left(\frac{1}{2}\right)\frac{1}{2\pi}\int_0^{2\pi} V_m^2 \sin^2(\omega t)\, d(\omega t)}$$

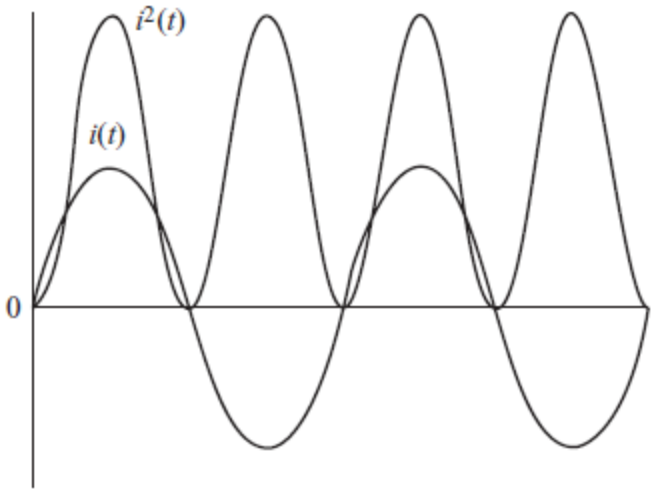
Taking the 1/2 outside of the square root gives

$$V_{\text{rms}} = \left(\sqrt{\frac{1}{2}}\right)\sqrt{\frac{1}{2\pi}\int_0^{2\pi} V_m^2 \sin^2(\omega t)\, d(\omega t)}$$

The last term on the right is the rms value of a sine wave which is known to be $V_m / \sqrt{2}$, so the rms value of a half-wave rectified sine wave is

$$V_{\text{rms}} = \sqrt{\frac{1}{2}}\,\frac{V_m}{\sqrt{2}} = \frac{V_m}{2}$$

Figure 2-7 shows the waveforms.



(*a*)

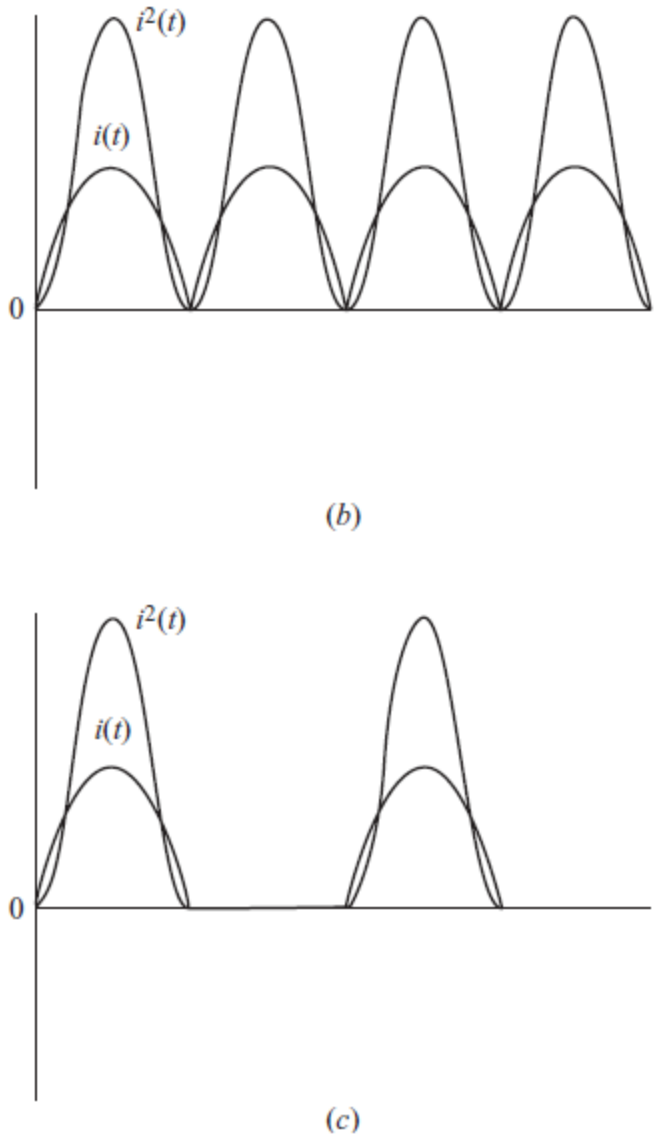


Figure 2-7 Waveforms and their squares for Example 2-5 (*a*) Sine wave; (*b*) full-wave rectified sine wave; (*c*) half-wave rectified sine wave.

EXAMPLE 2-6 (Neutral Conductor Current in a Three-Phase System)
An office complex is supplied from a three-phase four-wire voltage source (Fig. 2-8*a*).The load is highly nonlinear as a result of the rectifiers in the power supplies of the equipment, and the current in each of the three phases is shown in Fig. 2-8*b*. The neutral current is the sum of the phase currents. If the rms current in each phase conductor is known to be 20 A, determine the rms current in the neutral conductor.

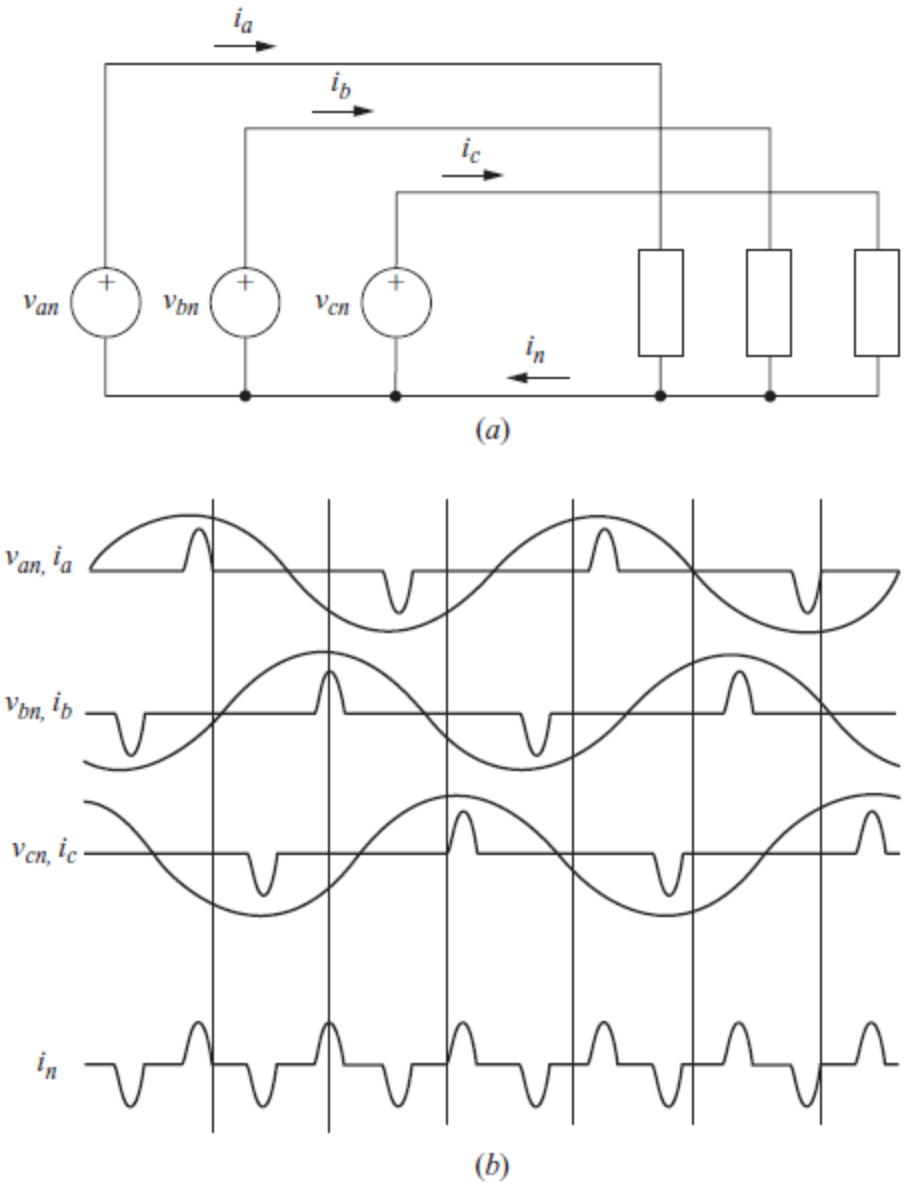
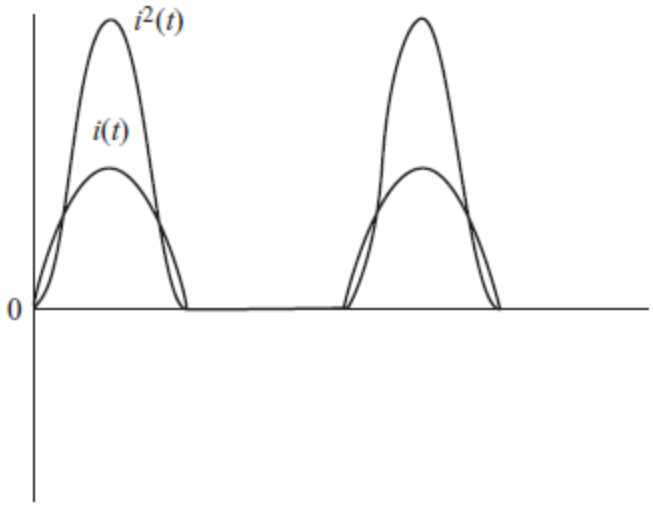


Figure 2-8 (*a*) Three-phase source supplying a balanced nonlinear three-phase load for Example 2-8; (*b*) phase and neutral currents;

■ Solution

Equation (2-38) may be applied to this case. Noting by inspection that the area of the square of the current function in the neutral *in*, is 3 times that of each of the phases i_a (Fig. 2-8*c*)

$$I_{n,\,\text{rms}} = \sqrt{\frac{1}{T}\int_0^T i_n^2(t)\,d(t)} = \sqrt{3\left(\frac{1}{T}\int_0^T i_a^2(t)\,d(t)\right)} = \sqrt{3}\,I_{a,\,\text{rms}}$$



(*c*)

Figure 2-8 (*c*) squares of i_a and i_n.

The rms current in the neutral is therefore:

$$I_{n,\,\text{rms}} = \sqrt{3}(20) = 34.6 \text{ A}$$

Note that the rms neutral current is larger than the phase currents for this situation.

This is much different from that for balanced linear loads, where the line currents are sinusoids that are displaced by 120° and sum to zero. Three-phase distribution systems supplying highly nonlinear loads should have a neutral conductor capable of carrying $\sqrt{3}$ times the current of the line conductors.

If a periodic voltage is the sum of two periodic voltage waveforms, $v(t) = v_1(t) + v_2(t)$, the rms value of $v(t)$ is determined from Eq. (2-37) as:

$$V_{\text{rms}}^2 = \frac{1}{T}\int_0^T (v_1 + v_2)^2\, dt = \frac{1}{T}\int_0^T \left(v_1^2 + 2v_1v_2 + v_2^2\right) dt$$

or

$$V_{\text{rms}}^2 = \frac{1}{T}\int_0^T v_1^2\, dt + \frac{1}{T}\int_0^T 2v_1v_2\, dt + \frac{1}{T}\int_0^T v_2^2\, dt$$

The term containing the product v_1v_2 in the above equation is zero if the functions v_1 and v_2 are orthogonal. A condition that satisfies that

requirement occurs when v_1 and v_2 are sinusoids of different frequencies. For orthogonal functions,

$$V_{\text{rms}}^2 = \frac{1}{T}\int_0^T v_1^2(t)\,dt + \frac{1}{T}\int_0^T v_2^2(t)\,dt$$

Noting that

$$\frac{1}{T}\int_0^T v_1^2(t)\,dt = V_{1,\text{rms}}^2 \quad \text{and} \quad \frac{1}{T}\int_0^T v_2^2(t)\,dt = V_{2,\text{rms}}^2$$

then

$$V_{\text{rms}} = \sqrt{V_{1,\text{rms}}^2 + V_{2,\text{rms}}^2}$$

If a voltage is the sum of more than two periodic voltages, all orthogonal, the rms value is

$$V_{\text{rms}} = \sqrt{V_{1,\text{rms}}^2 + V_{2,\text{rms}}^2 + V_{3,\text{rms}}^2 + \cdots} = \sqrt{\sum_{n=1}^{N} V_{n,\text{rms}}^2} \qquad (2\text{-}39)$$

Similarly,

$$I_{\text{rms}} = \sqrt{I_{1,\text{rms}}^2 + I_{2,\text{rms}}^2 + I_{3,\text{rms}}^2 + \cdots} = \sqrt{\sum_{n=1}^{N} I_{\text{n,rms}}^2} \qquad (2\text{-}40)$$

Note that Eq. (2-40) can be applied to Example 2-6 to obtain the rms value of the neutral current.

Example 2-7 (RMS Value of the Sum of Waveforms)
Determine the effective (rms) value of $v(t) = 4 + 8\sin(\omega_1 t + 10°) + 5\sin(\omega_2 t + 50°)$ for: (*a*) $\omega_2 = 2\,\omega_1$ and (*b*) $\omega_2 = \omega_1$.

■ **Solution**

(*a*) The rms value of a single sinusoid is $\frac{V_m}{\sqrt{2}}$ and the rms value of a constant is the constant. When the sinusoids are of different frequencies, the terms are orthogonal and Eq. (2-39) applies.

$$V_{\text{rms}} = \sqrt{V_{1,\text{rms}}^2 + V_{2,\text{rms}}^2 + V_{3,\text{rms}}^2} = \sqrt{4^2 + \left(\frac{8}{\sqrt{2}}\right)^2 + \left(\frac{5}{\sqrt{2}}\right)^2} = 7.78\ \text{V}$$

(*b*) For sinusoids of the same frequency, Eq. (2-39) does not apply because the integral of the cross product over one period is not zero. First combine the sinusoids using phasor addition:

$$8\angle 10^\circ + 5\angle 50^\circ = 12.3\angle 25.2^\circ$$

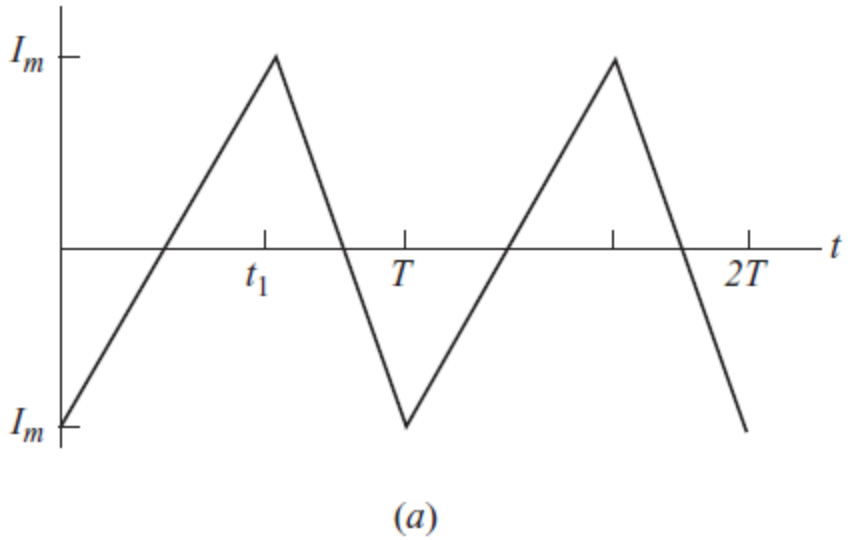
The voltage function is then expressed as

$$v(t) = 4 + 12.3 \sin(\omega_1 t + 25.2^\circ)\ \text{V}$$

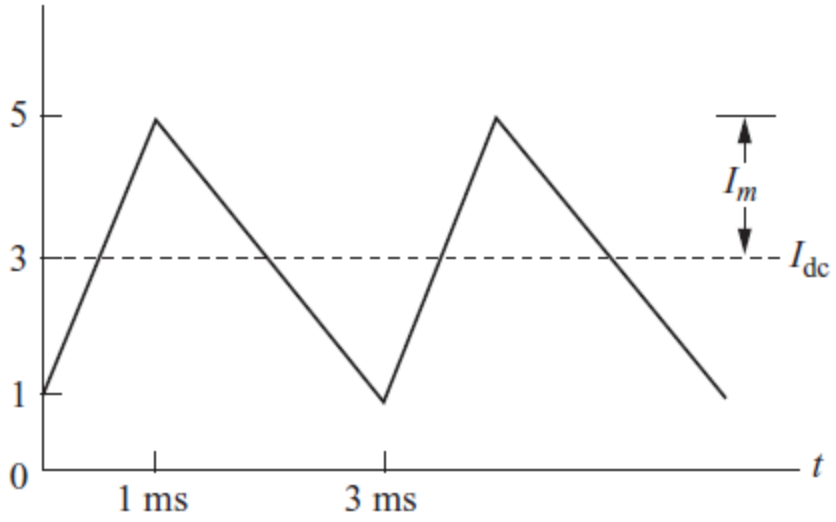
The rms value of this voltage is determined from Eq. (2-39) as

$$V_{\text{rms}} = \sqrt{4^2 + \left(\frac{12.3}{\sqrt{2}}\right)^2} = 9.57\ \text{V}$$

Example 2-8 (RMS Value of Triangular Waveforms)
(*a*) A triangular current waveform like that shown in Fig. 2-9a is commonly encountered in dc power supply circuits. Determine the rms value of this current.
(*b*) Determine the rms value of the offset triangular waveform in Fig. 2-9*b*.



(*a*)



(*b*)

Figure 2-9 (*a*) Triangular waveform for Example 2-8; (*b*) offset triangular waveform.

■ **Solution**

(*a*) The current is expressed as

$$i(t) = \begin{cases} \dfrac{2I_m}{t_1}t - I_m & 0 < t < t_1 \\ \dfrac{-2I_m}{T - t_1}t + \dfrac{I_m(T + t_1)}{T - t_1} & t_1 < t < T \end{cases}$$

The rms value is determined from Eq. (2-38).

$$I_{\text{rms}}^2 = \frac{1}{T}\left[\int_0^{t_1}\left(\frac{2I_m}{t_1}t - I_m\right)^2 dt + \int_{t_1}^{T}\left(\frac{-2I_m}{T-t_1}t + \frac{I_m(T+t_1)}{T-t_1}\right)^2 dt\right]$$

The details of the integration are quite long, but the result is simple: The rms value of a triangular current waveform is:

$$I_{\text{rms}} = \frac{I_m}{\sqrt{3}}$$

(*b*) The rms value of the offset triangular waveform can be determined by using the result of part (*a*). Since the triangular waveform of part (*a*) contains no dc component, the dc signal and the triangular waveform are orthogonal, and Eq. (2-40) applies.

$$I_{\text{rms}} = \sqrt{I_{1,\text{rms}}^2 + I_{2,\text{rms}}^2} = \sqrt{\left(\frac{I_m}{\sqrt{3}}\right)^2 + I_{\text{dc}}^2} = \sqrt{\left(\frac{2}{\sqrt{3}}\right)^2 + 3^2} = 3.22 \text{ A}$$

2.6 Apparent Power and Power Factor

Apparent Power *S*

Apparent power is the product of rms voltage and rms current magnitudes and is often used in specifying the rating of power equipment such as transformers.

Apparent power is expressed as:

$$\boxed{S = V_{\text{rms}} I_{\text{rms}}} \qquad (2\text{-}41)$$

In ac circuits (linear circuits with sinusoidal sources), apparent power is the magnitude of complex power.

Power Factor

The *power factor* of a load is defined as the ratio of average power to apparent power:

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{\text{rms}} I_{\text{rms}}} \qquad (2\text{-}42)$$

In sinusoidal ac circuits, the above calculation results in $pf = \cos\theta$ where θ is the phase angle between the voltage and current sinusoids. However, that is a special case and should be used only when both voltage and current are sinusoids. In general, power factor must be computed from Eq. (2-42).

2.7 Power Computations for Sinusoidal AC Circuits

In general, voltages and/or currents in power electronics circuits are not sinusoidal.

However, a nonsinusoidal periodic waveform can be represented by a Fourier series of sinusoids. It is therefore important to understand thoroughly power computations for the sinusoidal case. The following discussion is a review of power computations for ac circuits.

For linear circuits that have sinusoidal sources, all steady-state voltages and currents are sinusoids. Instantaneous power and average power for ac circuits are computed using Eqs. (2-1) and (2-3) as follows:

For any element in an ac circuit, let

$$v(t) = V_m \cos(\omega t + \theta)$$
$$i(t) = I_m \cos(\omega t + \phi) \qquad (2\text{-}43)$$

Then instantaneous power is

$$p(t) = v(t)i(t) = [V_m \cos(\omega t + \theta)][I_m \cos(\omega t + \phi)] \qquad (2\text{-}44)$$

Using the trigonometric identity gives

$$(\cos A)(\cos B) = \frac{1}{2}[\cos(A + B) + \cos(A - B)] \qquad (2\text{-}45)$$

$$p(t) = \left(\frac{V_m I_m}{2}\right)[\cos(2\omega t + \theta + \phi) + \cos(\theta - \phi)] \qquad (2\text{-}46)$$

Average power is

$$P = \frac{1}{T}\int_0^T p(t)\,dt = \left(\frac{V_m I_m}{2}\right)\int_0^T [\cos(2\omega t + \theta + \phi) + \cos(\theta - \phi)]\,dt \qquad (2\text{-}47)$$

The result of the above integration can be obtained by inspection. Since the first term in the integration is a cosine function, the integral over one period is zero because of equal areas above and below the time axis. The second term in the integration is the constant $cos(\theta - \emptyset)$, which has an average value of $cos(\theta - \emptyset)$.

Therefore, the average power in any element in an ac circuit is:

$$P = \left(\frac{V_m I_m}{2}\right)\cos(\theta - \phi) \qquad (2\text{-}48)$$

This equation is frequently expressed as

$$P = V_{\text{rms}} I_{\text{rms}} \cos(\theta - \phi) \qquad (2\text{-}49)$$

Where $V_{rms} = V_m/\sqrt{2}$, $I_{rms} = I_m/\sqrt{2}$, and $\theta - \emptyset$ is the phase angle between voltage and current. The power factor is determined to be $cos\ (\theta - \emptyset\ )$ by using Eq. (2-42).

In the steady state, no net power is absorbed by an inductor or a capacitor. The term *reactive power* is commonly used in conjunction with voltages and currents for inductors and capacitors. Reactive power is characterized by energy storage during one-half of the cycle and energy retrieval during the other half.

Reactive power is computed with a relationship similar to Eq. (2-49):

$$Q = V_{\text{rms}} I_{\text{rms}} \sin(\theta - \phi) \qquad (2\text{-}50)$$

By convention, inductors absorb positive reactive power and capacitors absorb negative reactive power.

Complex power combines real and reactive powers for ac circuits:

$$\mathbf{S} = P + jQ = (\mathbf{V}_{\text{rms}})(\mathbf{I}_{\text{rms}})^* \qquad (2\text{-}51)$$

In the above equation, $\boldsymbol{V}_{rms}$ and $\boldsymbol{I}_{rms}$ are complex quantities often expressed as phasors (magnitude and angle), and $(\boldsymbol{I}_{rms})^*$ is the complex conjugate of phasor current, which gives results consistent with the convention that inductance, or lagging current, absorbs reactive power.

Apparent power in AC circuits is the magnitude of complex power:

$$S = |\mathbf{S}| = \sqrt{P^2 + Q^2} \qquad (2\text{-}52)$$

It is important to note that the complex power in Eq. (2-52) and power factor of $cos\ (\theta - \emptyset\ )$ for sinusoidal ac circuits are special cases and are not applicable to nonsinusoidal voltages and currents.

2.8 POWER COMPUTATIONS FOR NONSINUSOIDAL PERIODIC WAVEFORMS

Power electronics circuits typically have voltages and/or currents that are periodic but not sinusoidal. For the general case, the basic definitions for the power terms described at the beginning of this chapter must be applied.

Acommon error that is made when doing power computations is to attempt to apply some special relationships for sinusoids to waveforms that are not sinusoids.

The Fourier series can be used to describe non-sinusoidal periodic waveforms in terms of a series of sinusoids. The power relationships for

these circuits can be expressed in terms of the components of the Fourier series.

Fourier Series

A non-sinusoidal periodic waveform that meets certain conditions can be described by a Fourier series of sinusoids. The Fourier series for a periodic function $f(t)$ can be expressed in trigonometric form as:

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] \tag{2-53}$$

Where:

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t)\, dt$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(n\omega_0 t)\, dt \tag{2-54}$$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(n\omega_0 t)\, dt$$

Sines and cosines of the same frequency can be combined into one sinusoid, resulting in an alternative expression for a Fourier series:

$$f(t) = a_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

where (2-55)

$$C_n = \sqrt{a_n^2 + b_n^2} \quad \text{and} \quad \theta_n = \tan^{-1}\left(\frac{-b_n}{a_n}\right)$$

or

$$f(t) = a_0 + \sum_{n=1}^{\infty} C_n \sin(n\omega_0 t + \theta_n)$$

where (2-56)

$$C_n = \sqrt{a_n^2 + b_n^2} \quad \text{and} \quad \theta_n = \tan^{-1}\left(\frac{a_n}{b_n}\right)$$

The term a_0 is a constant that is the average value of $f(t)$ and represents a DC voltage or current in electrical applications. The coefficient C_1 is the amplitude of the term at the fundamental frequency ω_0. Coefficients $C_2, C_3, \ldots$ are the amplitudes of the harmonics that have frequencies $2\,\omega_0, 3\,\omega_0, \ldots$.

The rms value of $f(t)$ can be computed from the Fourier series:

$$F_{\text{rms}} = \sqrt{\sum_{n=0}^{\infty} F_{n,\text{rms}}^2} = \sqrt{a_0^2 + \sum_{n=1}^{\infty}\left(\frac{C_n}{\sqrt{2}}\right)^2} \qquad (2\text{-}57)$$

Average Power

If periodic voltage and current waveforms represented by the Fourier series

$$v(t) = V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega_0 t + \theta_n)$$
$$i(t) = I_0 + \sum_{n=1}^{\infty} I_n \cos(n\omega_0 t + \phi_n) \qquad (2\text{-}58)$$

exist for a device or circuit, then average power is computed from Eq. (2-3).

$$P = \frac{1}{T}\int_0^T v(t)i(t)\,dt$$

The average of the products of the dc terms is $V_0\ I_0$. The average of voltage and current products at the same frequency is described by Eq. (2-49), and the average of voltage and current products of different frequencies is zero. Consequently, average power for non-sinusoidal periodic voltage and current waveforms is:

$$P = \sum_{n=0}^{\infty} P_n = V_0 I_0 + \sum_{n=1}^{\infty} V_{n,\,\text{rms}} I_{n,\,\text{rms}} \cos(\theta_n - \phi_n)$$

or

$$P = V_0 I_0 + \sum_{n=1}^{\infty} \left(\frac{V_{n,\,\max} I_{n,\,\max}}{2} \right) \cos(\theta_n - \phi_n) \qquad (2\text{-}59)$$

Note that total average power is the sum of the powers at the frequencies in the Fourier series.

Nonsinusoidal Source and Linear Load

If a nonsinusoidal periodic voltage is applied to a load that is a combination of linear elements, the power absorbed by the load can be determined by using superposition. A nonsinusoidal periodic voltage is equivalent to the series combination of the Fourier series voltages, as illustrated in Fig. 2-10. The current in the load can be determined using superposition, and Eq. (2-59) can be applied to compute average power. Recall that superposition for power is not valid when the sources are of the same frequency. The technique is demonstrated in Example 2-9.

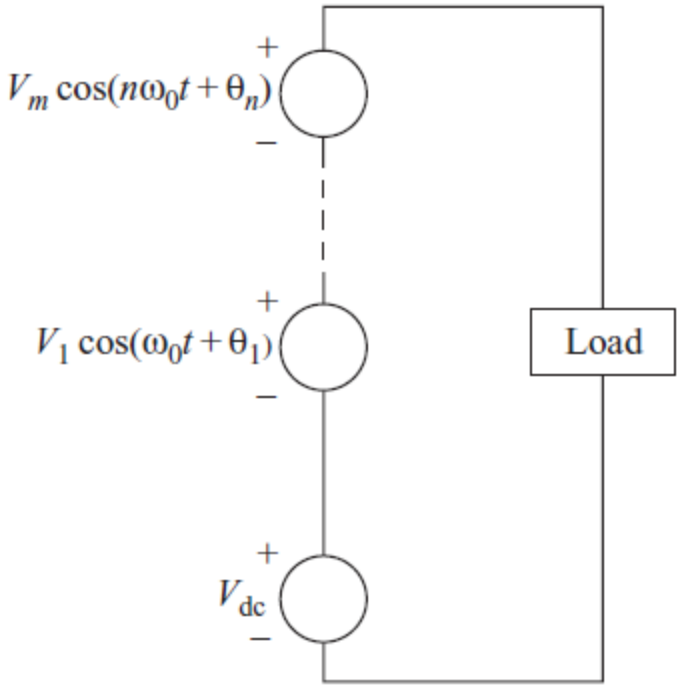


Figure 2.10 Equivalent circuit for Fourier analysis.

Example 2-9 (Nonsinusoidal Source and Linear Load)
A non-sinusoidal periodic voltage has a Fourier series of $v(t) = 10 + 20\cos(2\pi 60t - 25°) + 30\cos(4\pi 60t + 20°)$ V. This voltage is connected to a load that is a 5 Ω resistor and a 15-mH inductor connected in series as in Fig. 2-11. Determine the power absorbed by the load.

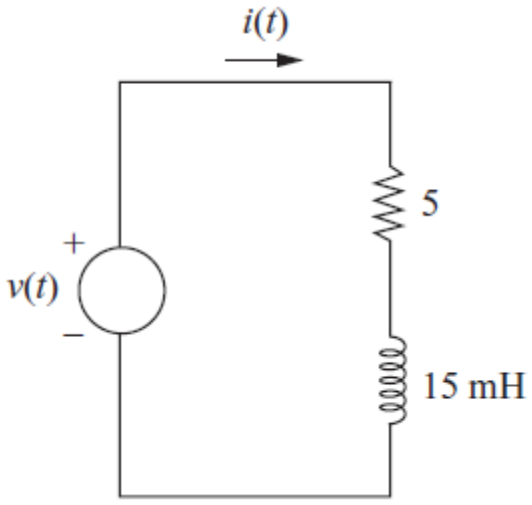


Figure 2.11 Circuit for Example 2-9.

■ **Solution**

Current at each source frequency is computed separately. The DC current term is:

$$I_0 = \frac{V_0}{R} = \frac{10}{5} = 2 \text{ A}$$

The amplitudes of the AC current terms are computed from phasor analysis:

$$\boldsymbol{I}_1 = \frac{\boldsymbol{V}_1}{R + j\omega_1 L} = \frac{20\angle(-25°)}{5 + j(2\pi 60)(0.015)} = 2.65\angle(-73.5°) \text{ A}$$

$$\boldsymbol{I}_2 = \frac{\boldsymbol{V}_2}{R + j\omega_2 L} = \frac{30\angle 20°}{5 + j(4\pi 60)(0.015)} = 2.43\angle(-46.2°) \text{ A}$$

Load current can then be expressed as

$$i(t) = 2 + 2.65\cos(2\pi 60t - 73.5°) + 2.43\cos(4\pi 60t - 46.2°) \text{ A}$$

Power at each frequency in the Fourier series is determined from Eq. (2-59):

$$\text{dc term:} \qquad P_0 = (10 \text{ V})(2 \text{ A}) = 20 \text{ W}$$

$$\omega = 2\pi 60: \qquad P_1 = \frac{(20)(2.65)}{2}\cos(-25° + 73.5°) = 17.4 \text{ W}$$

$$\omega = 4\pi 60: \qquad P_2 = \frac{(30)(2.43)}{2}\cos(20° + 46°) = 14.8 \text{ W}$$

Total power is then

$$P = 20 + 17.4 + 14.8 = 52.2 \text{ W}$$

Power absorbed by the load can also be computed from $I_{rms}^2 R$ in this circuit because the average power in the inductor is zero.

$$P = I_{\text{rms}}^2 R = \left[2^2 + \left(\frac{2.65}{\sqrt{2}}\right)^2 + \left(\frac{2.43}{\sqrt{2}}\right)^2\right]5 = 52.2 \text{ W}$$

Sinusoidal Source and Nonlinear Load

If a sinusoidal voltage source is applied to a nonlinear load, the current waveform will not be sinusoidal but can be represented as a Fourier series. If voltage is the sinusoid

$$v(t) = V_1 \sin(\omega_0 t + \theta_1) \tag{2-60}$$

and current is represented by the Fourier series

$$i(t) = I_0 + \sum_{n=1}^{\infty} I_n \sin(n\omega_0 t + \phi_n) \tag{2-61}$$

then average power absorbed by the load (or supplied by the source) is computed from Eq. (2-59) as

$$\begin{aligned} P &= V_0 I_0 + \sum_{n=1}^{\infty} \left(\frac{V_{n,\max} I_{n,\max}}{2} \right) \cos(\theta_n - \phi_n) \\ &= (0)(I_0) + \left(\frac{V_1 I_1}{2} \right) \cos(\theta_1 - \phi_1) + \sum_{n=2}^{\infty} \frac{(0)(I_{n,\max})}{2} \cos(\theta_n - \phi_n) \\ &= \left(\frac{V_1 I_1}{2} \right) \cos(\theta_1 - \phi_1) = V_{1,\text{rms}} I_{1,\text{rms}} \cos(\theta_1 - \phi_1) \end{aligned} \tag{2-62}$$

Note that the only nonzero power term is at the frequency of the applied voltage.

The power factor of the load is computed from Eq. (2-42).

$$\begin{aligned} \text{pf} &= \frac{P}{S} = \frac{P}{V_{\text{rms}} I_{\text{rms}}} \\ \text{pf} &= \frac{V_{1,\text{rms}} I_{1,\text{rms}} \cos(\theta_1 - \phi_1)}{V_{1,\text{rms}} I_{\text{rms}}} = \left(\frac{I_{1,\text{rms}}}{I_{\text{rms}}} \right) \cos(\theta_1 - \phi_1) \end{aligned} \tag{2-63}$$

Where rms current is computed from

$$I_{\text{rms}} = \sqrt{\sum_{n=0}^{\infty} I_{n,\text{rms}}^2} = \sqrt{I_0^2 + \sum_{n=1}^{\infty} \left(\frac{I_n}{\sqrt{2}} \right)^2} \tag{2-64}$$

Note also that for a sinusoidal voltage and a sinusoidal current, $pf = cos(\theta_1 - \emptyset_1)$), which is the power factor term commonly used in linear circuits and is called the *displacement power factor*. The ratio of the rms value of the fundamental frequency to the total rms value, $I_{1,rms}/I_{rms}$ in Eq. (2-63), is the *distortion factor* (DF).

$$\text{DF} = \frac{I_{1,\text{rms}}}{I_{\text{rms}}} \tag{2-65}$$

The distortion factor represents the reduction in power factor due to the nonsinusoidal property of the current. Power factor is also expressed as

$$\text{pf} = [\cos(\theta_1 - \phi_1)]\,\text{DF} \tag{2-66}$$

Total harmonic distortion (THD) is another term used to quantify the nonsinusoidal property of a waveform. THD is the ratio of the rms value of all the nonfundamental frequency terms to the rms value of the fundamental frequency term.

$$\text{THD} = \sqrt{\frac{\sum_{n\neq 1} I_{n,\text{rms}}^2}{I_{1,\text{rms}}^2}} = \frac{\sqrt{\sum_{n\neq 1} I_{n,\text{rms}}^2}}{I_{1,\text{rms}}} \tag{2-67}$$

THD is equivalently expressed as

$$\text{THD} = \sqrt{\frac{I_{\text{rms}}^2 - I_{1,\text{rms}}^2}{I_{1,\text{rms}}^2}} \tag{2-68}$$

Total harmonic distortion is often applied in situations where the dc term is zero, in which case THD may be expressed as

$$\text{THD} = \frac{\sqrt{\sum_{n=2}^{\infty} I_n^2}}{I_1} \tag{2-69}$$

Another way to express the distortion factor is

$$\text{DF} = \sqrt{\frac{1}{1 + (\text{THD})^2}} \tag{2-70}$$

Reactive power for a sinusoidal voltage and a nonsinusoidal current can be expressed as in Eq. (2-50). The only nonzero term for reactive power is at the voltage frequency:

$$Q = \frac{V_1 I_1}{2} \sin(\theta_1 - \phi_1) \tag{2-71}$$

With P and Q defined for the nonsinusoidal case, apparent power S must include a term to account for the current at frequencies which are different

from the voltage frequency. The term distortion volt-amps D is traditionally used in the computation of S,

$$S = \sqrt{P^2 + Q^2 + D^2} \tag{2-72}$$

Where

$$D = V_{1,\,\text{rms}}\sqrt{\sum_{n\neq 1}^{\infty} I_{n,\text{rms}}^2} = \frac{V_1}{2}\sqrt{\sum_{n\neq 1}^{\infty} I_n} \tag{2-73}$$

Other terms that are sometimes used for nonsinusoidal current (or voltages) are form factor and crest factor.

$$\text{Form factor} = \frac{I_{\text{rms}}}{I_{\text{avg}}} \tag{2-74}$$

$$\text{Crest factor} = \frac{I_{\text{peak}}}{I_{\text{rms}}} \tag{2-75}$$

Example 2-10 (Sinusoidal Source and a Nonlinear Load)
A sinusoidal voltage source of $v(t) = 100\ cos(377t)\ V$ is applied to a nonlinear load, resulting in a nonsinusoidal current which is expressed in Fourier series form as

$$i(t) = 8 + 15\cos(377t + 30°) + 6\cos[2(377)t + 45°] + 2\cos[3(377)t + 60°]$$

Determine (*a*) the power absorbed by the load, (*b*) the power factor of the load, (*c*) the distortion factor of the load current, (*d*) the total harmonic distortion of the load current.

■ **Solution**

(*a*) The power absorbed by the load is determined by computing the power absorbed at each frequency in the Fourier series [Eq. (2-59)].

$$P = (0)(8) + \left(\frac{100}{\sqrt{2}}\right)\left(\frac{15}{\sqrt{2}}\right)\cos 30° + (0)\left(\frac{6}{\sqrt{2}}\right)\cos 45° + (0)\left(\frac{2}{\sqrt{2}}\right)\cos 60°$$

$$P = \left(\frac{100}{\sqrt{2}}\right)\left(\frac{15}{\sqrt{2}}\right)\cos 30° = 650\ \text{W}$$

(*b*) The rms voltage is

$$V_{\text{rms}} = \frac{100}{\sqrt{2}} = 70.7 \text{ V}$$

and the rms current is computed from Eq. (2-64):

$$I_{\text{rms}} = \sqrt{8^2 + \left(\frac{15}{\sqrt{2}}\right)^2 + \left(\frac{6}{\sqrt{2}}\right)^2 + \left(\frac{2}{\sqrt{2}}\right)^2} = 14.0 \text{ A}$$

The power factor is

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{\text{rms}} I_{\text{rms}}} = \frac{650}{(70.7)(14.0)} = 0.66$$

Alternatively, power factor can be computed from Eq. (2-63):

$$\text{pf} = \frac{I_{1,\text{ rms}} \cos(\theta_1 - \phi_1)}{I_{\text{rms}}} = \frac{\left(\frac{15}{\sqrt{2}}\right)\cos(0 - 30^\circ)}{14.0} = 0.66$$

(c) The distortion factor is computed from Eq. (2-65) as

$$\text{DF} = \frac{I_{1,\text{ rms}}}{I_{\text{rms}}} = \frac{\frac{15}{\sqrt{2}}}{14.0} = 0.76$$

(d) The total harmonic distortion of the load current is obtained from Eq. (2-68).

$$\text{THD} = \sqrt{\frac{I_{\text{rms}}^2 - I_{1,\text{rms}}^2}{I_{1,\text{rms}}^2}} = \sqrt{\frac{14^2 - \left(\frac{15}{\sqrt{2}}\right)^2}{\left(\frac{15}{\sqrt{2}}\right)^2}} = 0.86 = 86\%.$$

2.9 Summary

• Instantaneous power is the product of voltage and current at a particular time:

$$p(t) = v(t)i(t)$$

Using the passive sign convention, the device is absorbing power if $(p)(t)$ is positive, and the device is supplying power if $(p)(t)$ is negative.

• Power usually refers to average power, which is the time average of periodic instantaneous power:

$$P = \frac{1}{T}\int_{t_0}^{t_0+T} v(t)i(t)\,dt = \frac{1}{T}\int_{t_0}^{t_0+T} p(t)\,dt$$

The rms value is the root-mean-square or effective value of a voltage or current waveform.

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}$$

$$I_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T i^2(t)\,dt}$$

Apparent power is the product of rms voltage and current.

$$S = V_{\text{rms}} I_{\text{rms}}$$

Power factor is the ratio of average power to apparent power.

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{\text{rms}} I_{\text{rms}}}$$

For inductors and capacitors that have periodic voltages and currents, the average power is zero. Instantaneous power is generally not zero because the device stores energy and then returns energy to the circuit.
• For periodic currents, the average voltage across an inductor is zero.
• For periodic voltages, the average current in a capacitor is zero.
• For nonsinusoidal periodic waveforms, average power may be computed from the basic definition, or the Fourier series method may be used. The Fourier series method treats each frequency in the series separately and uses superposition to compute total power.

$$P = \sum_{n=0}^{\infty} P_n = V_0 I_0 + \sum_{n=1}^{\infty} V_{n,\,\text{rms}}\, I_{n,\,\text{rms}} \cos\left(\theta_n - \phi_n\right)$$